Introduction to Data Science

Agenda

- What is Data?
- What is Data Science?
- Why are we discussing Data Science?
- Data Life Cycle
- Applications of Data Science

What is Data?

Data refers to raw, unprocessed facts, figures, or information that can be in various forms and formats. Data is the fundamental building block of information and knowledge, and it can represent a wide range of observations, measurements, or descriptions of objects, events, or phenomena.

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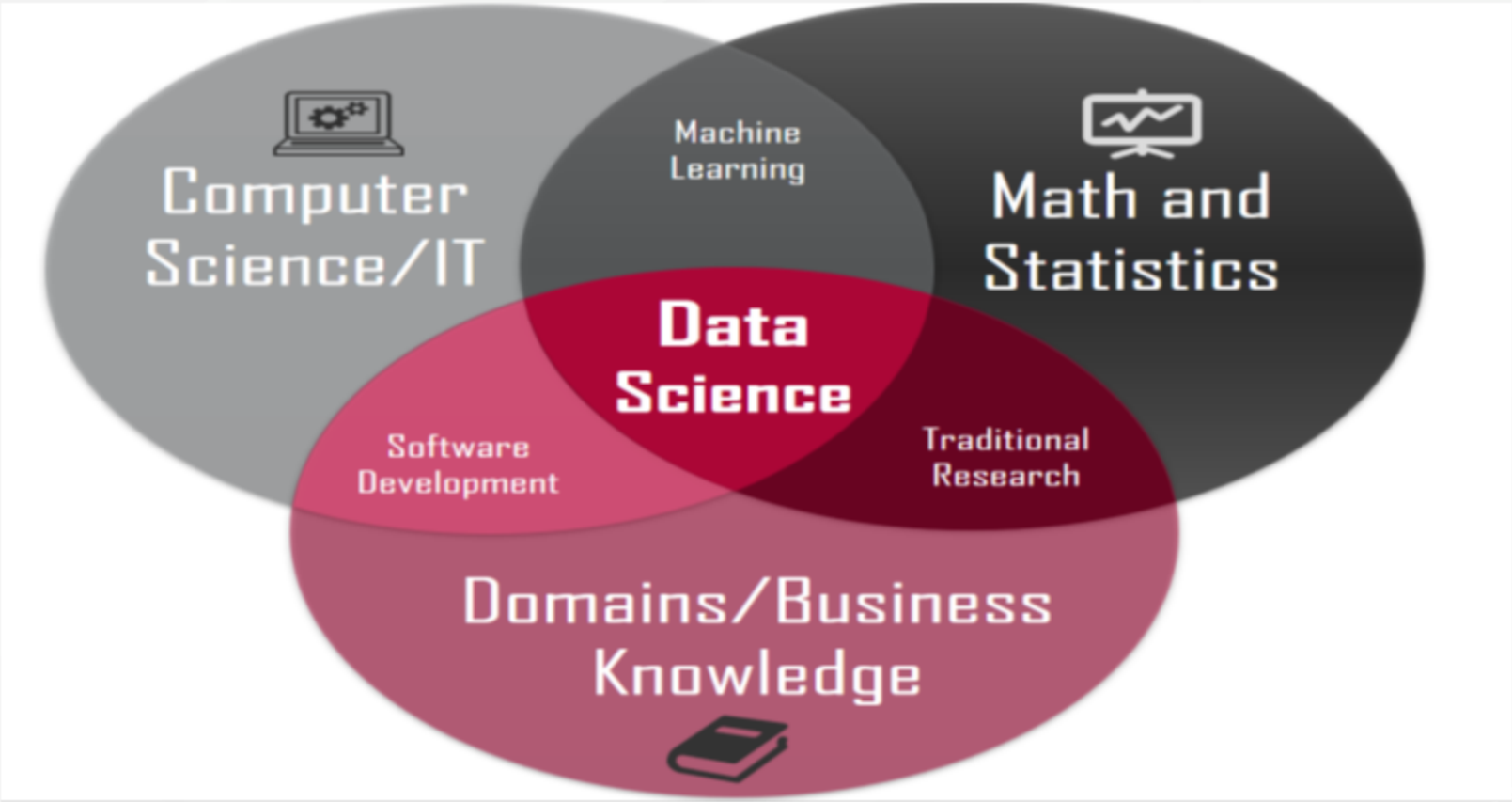
What is Data Science?

Data science is a multidisciplinary field that combines techniques and methods from various domains, including statistics, computer science, machine learning, and domain-specific knowledge, to extract insights and knowledge from data.



Why are we discussing Data Science?

Data science is increasingly important in today's world for several reasons, and its significance continues to grow.

- Explosion of Data
- Data Driven Decision Making
- Competitive Advantage
- Predictive Analytics
- Scientific Research
- Healthcare & Medicine
- Environmental Conservation

Data Life Cycle

- Data Collection
- Data Cleaning & Preprocessing
- Exploratory Data Analysis
- Feature Engineering
- Machine Learning
- Data Visualization
- Statistical Analysis
- Reporting & Documentation
- Iterative Process

Applications of Data Science

- Business Analytics:
  1. Market Segmentation
  2. Customer Churn Prediction
  3. Sales Forecasting
- Healthcare:
  1. Disease Diagnosis
  2. Drug Discovery
  3. Personalized Medicine
- Finance:
  1. Risk Assessment
  2. Portfolio Management
- Retail & E-commerce:
  1. Recommendation System
  2. Inventory Management

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